Exploratory Data Analysis (EDA) – Titanic Dataset

Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset to extract insights using visual and statistical exploration techniques.

Tools Used

• Python
• Pandas
• Matplotlib
• Seaborn

Steps and Key Insights

1. Data Loading and Preprocessing:
- Loaded the Titanic dataset using Pandas.
- Handled missing values and converted categorical variables where necessary.

2. Univariate Analysis:
- Analyzed distributions of individual variables like 'Age', 'Fare', and 'Survived'.
- Found that younger passengers and those who paid higher fares were more likely to survive.

3. Bivariate Analysis:
- Explored relationships between 'Survived' and other variables such as 'Sex', 'Pclass', 'SibSp', and 'Parch'.
- Females and first-class passengers had higher survival rates.
- Passengers with 1–2 family members had better chances of survival.

4. Multivariate & Correlation Analysis:
- Used heatmaps and pairplots to discover deeper patterns and correlations between numerical features.
- Fare and Pclass were moderately correlated.
- Pairplots showed distinct clusters based on survival, age, and class.

Conclusion

The EDA on the Titanic dataset highlighted several important trends and patterns:
- Gender was a strong predictor of survival; females were more likely to survive.

- First-class passengers had higher survival rates compared to second and third-class passengers.
- Children had a better chance of survival than adults.
- Passengers traveling alone or with large families had lower survival rates.
- Fare and passenger class were significantly correlated, impacting survival odds.

These insights provide a clear understanding of the factors that influenced survival during the Titanic disaster, and also demonstrate how data visualization and statistical tools can uncover meaningful patterns in real-world datasets.